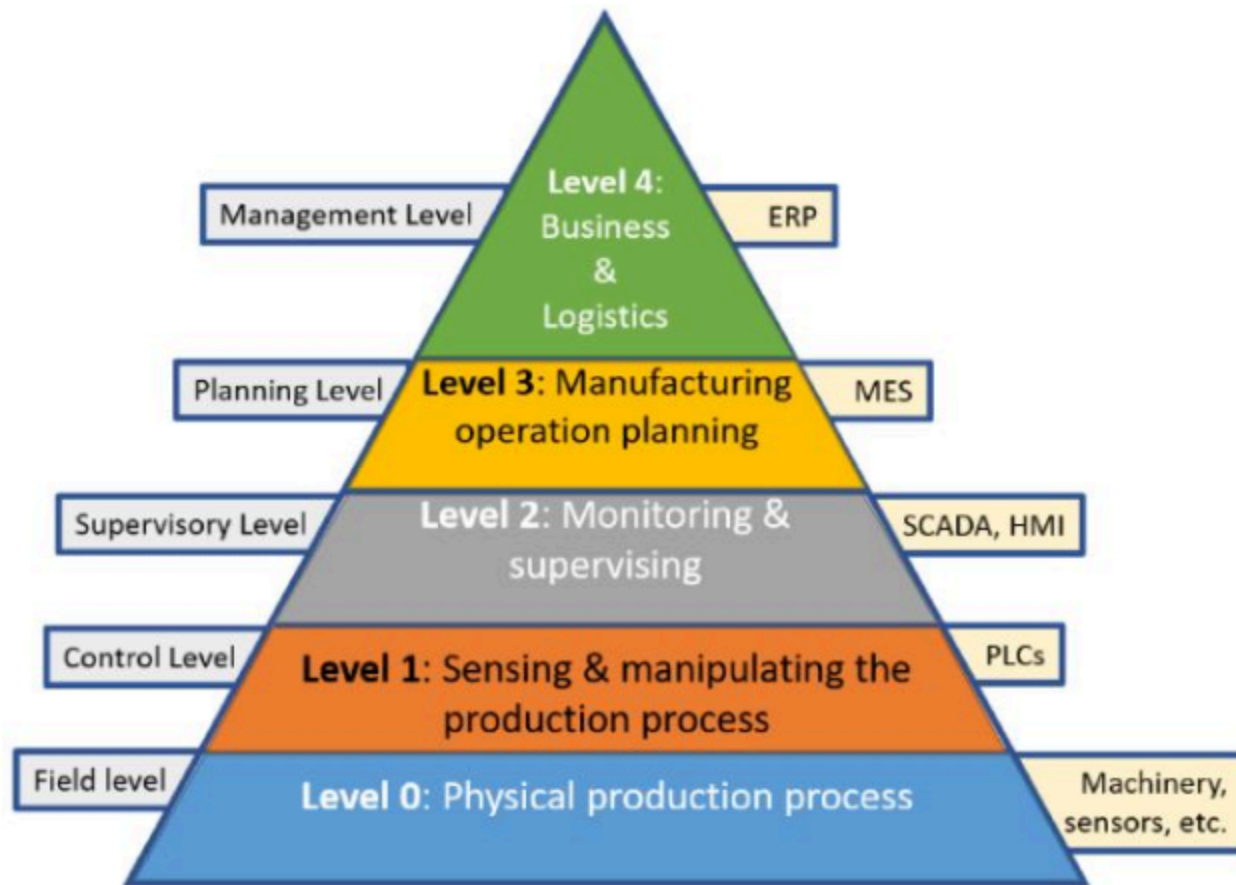


Production Supporting Systems in Factories

Smart Sensors and Information Systems

อุปกรณ์ตรวจวัดอัจฉริยะและระบบสารสนเทศ

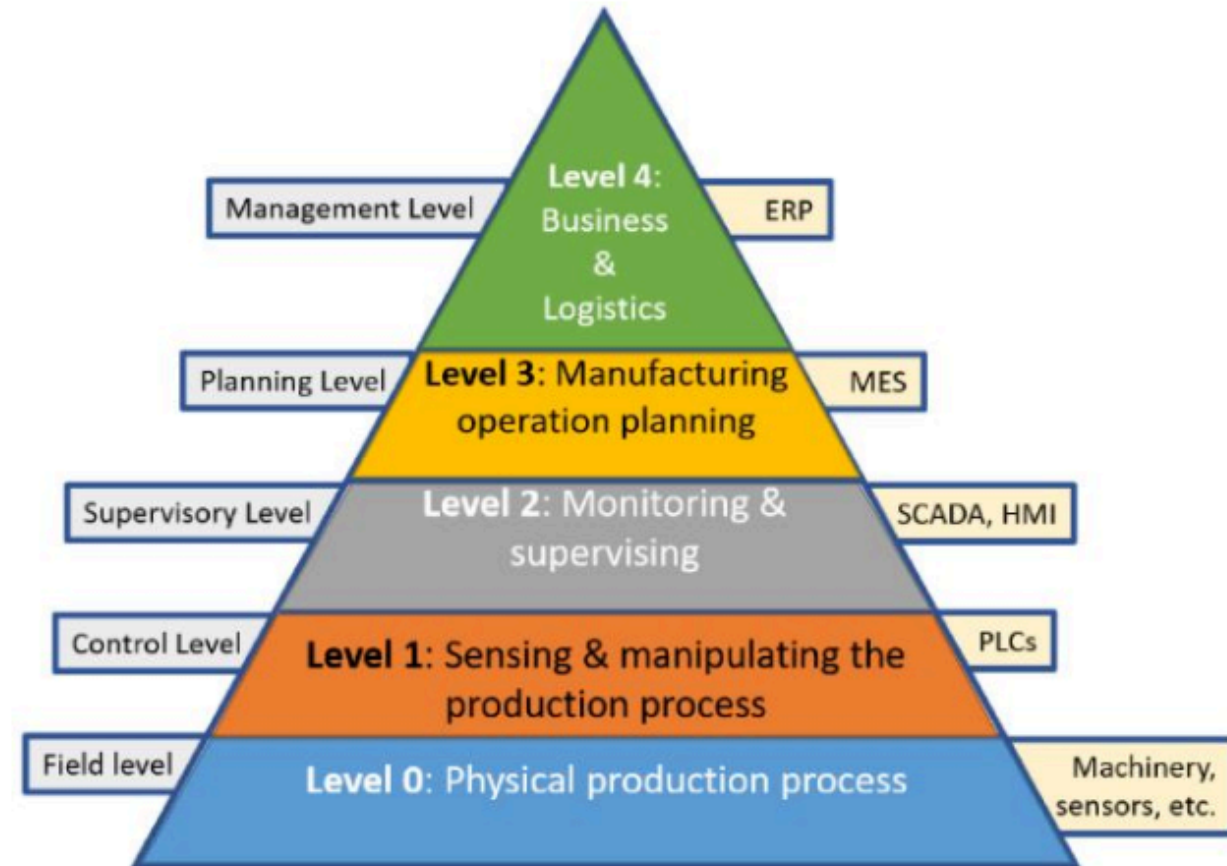


ISA-95 Standard

- International standard for the integration of enterprise and control systems.
- Defines the hierarchy of systems in manufacturing operations.

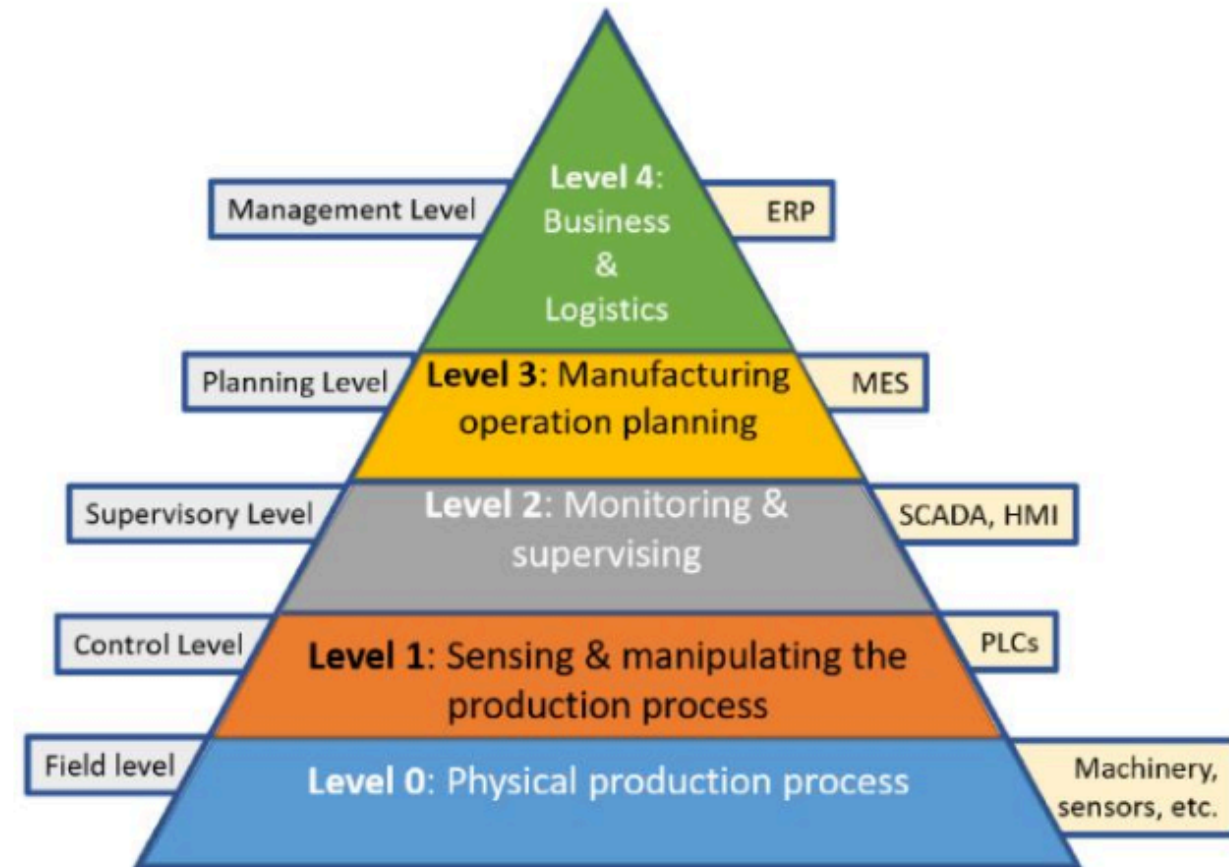
Level 0

- **Production Process**
- The actual physical and chemical processes on the plant floor.



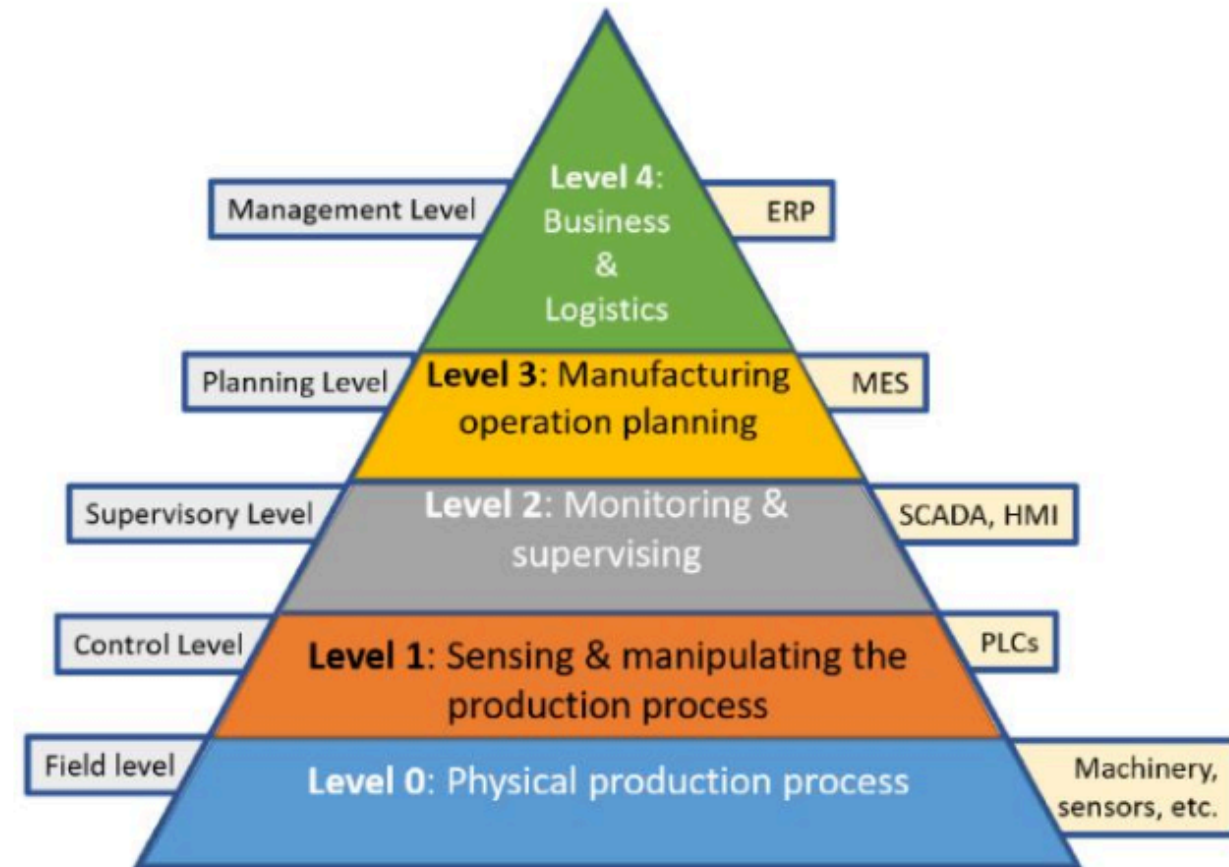
Level 1

- **Sensing and Manipulation**
- Sensors and actuators that interact directly with the production process.



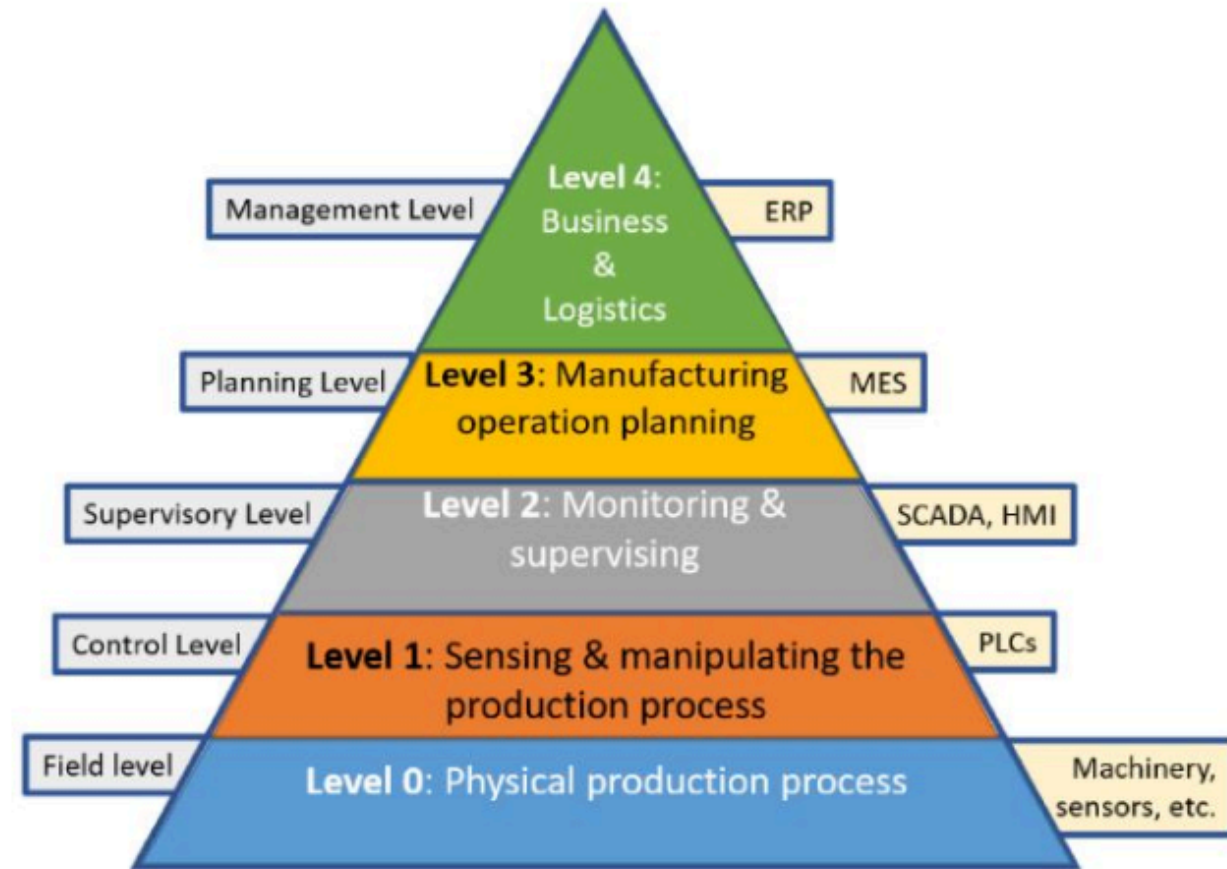
Levels 2

- **Monitoring and Supervision**
- Systems like SCADA and HMI that monitor and control the process in real-time.



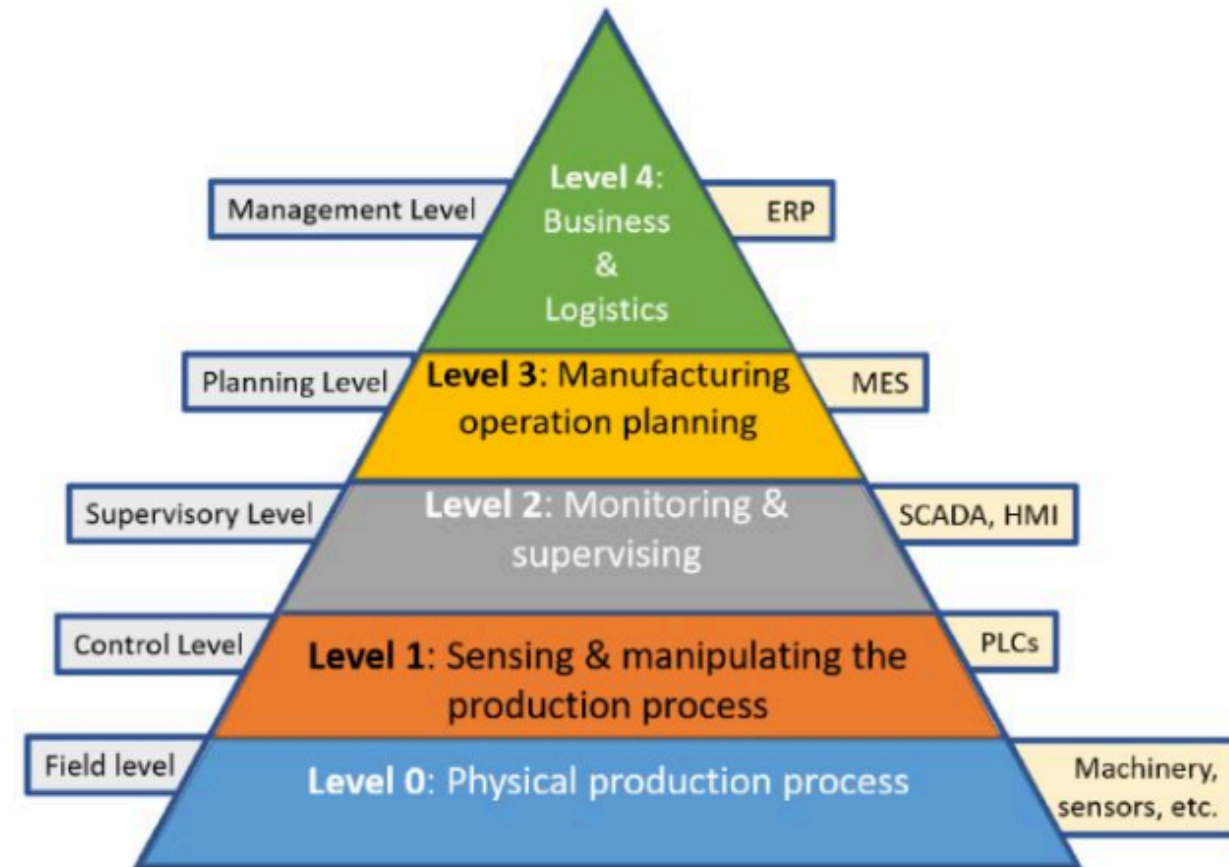
Levels 3

- **Manufacturing Operations Management**
- Manages day-to-day production, including scheduling, quality, and inventory. MES (Manufacturing Execution System) is a key component here.



Levels 4

- **Business Planning and Logistics**
- Handles enterprise-level functions like production planning, order processing, and supply chain management, typically managed by ERP systems.

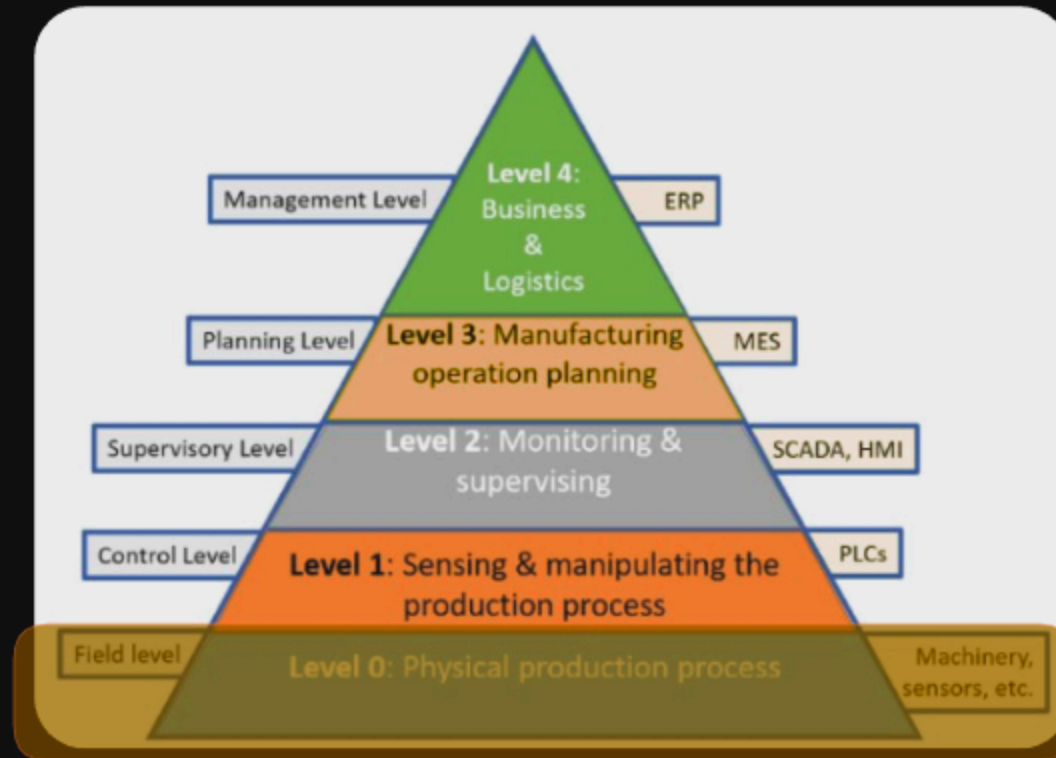


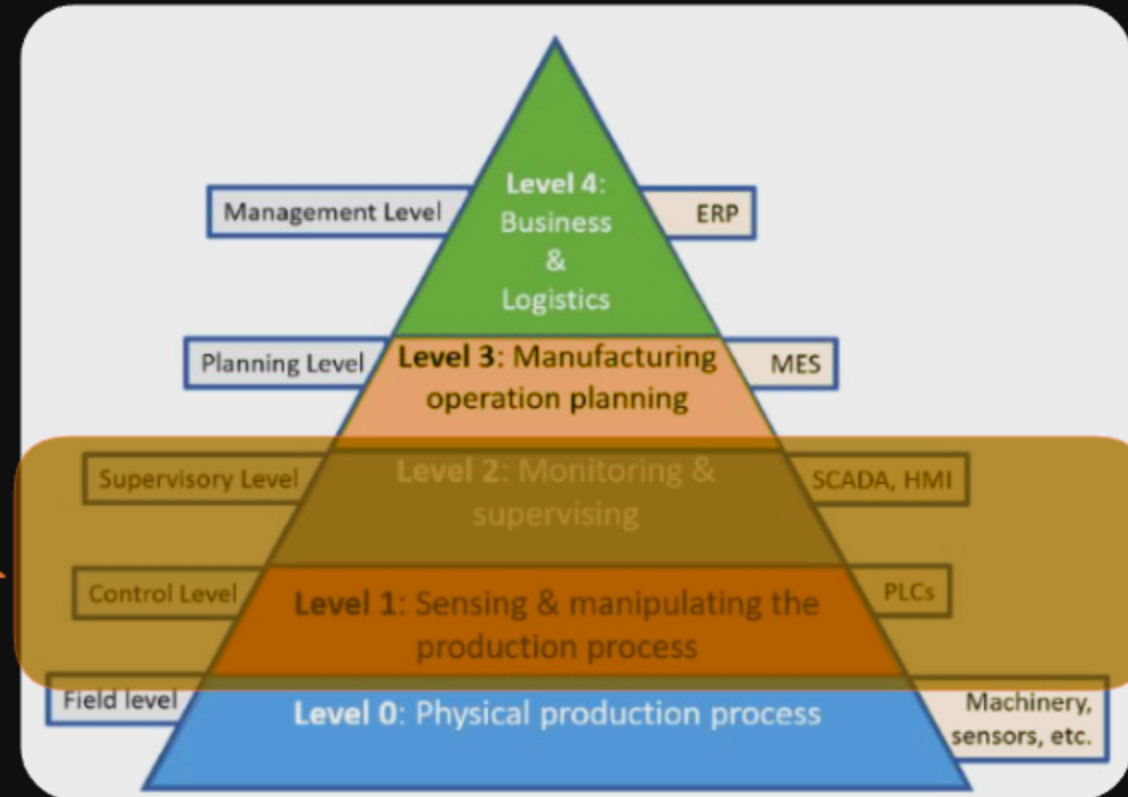
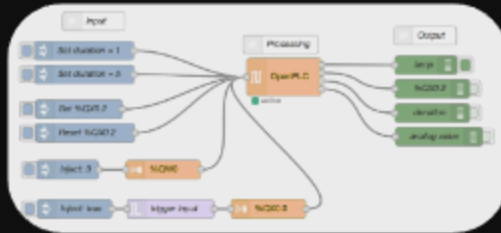
Note

- This diagram is rather conceptual and may not represent all real-world scenarios.
- In practice, systems can overlap and interact in complex ways.

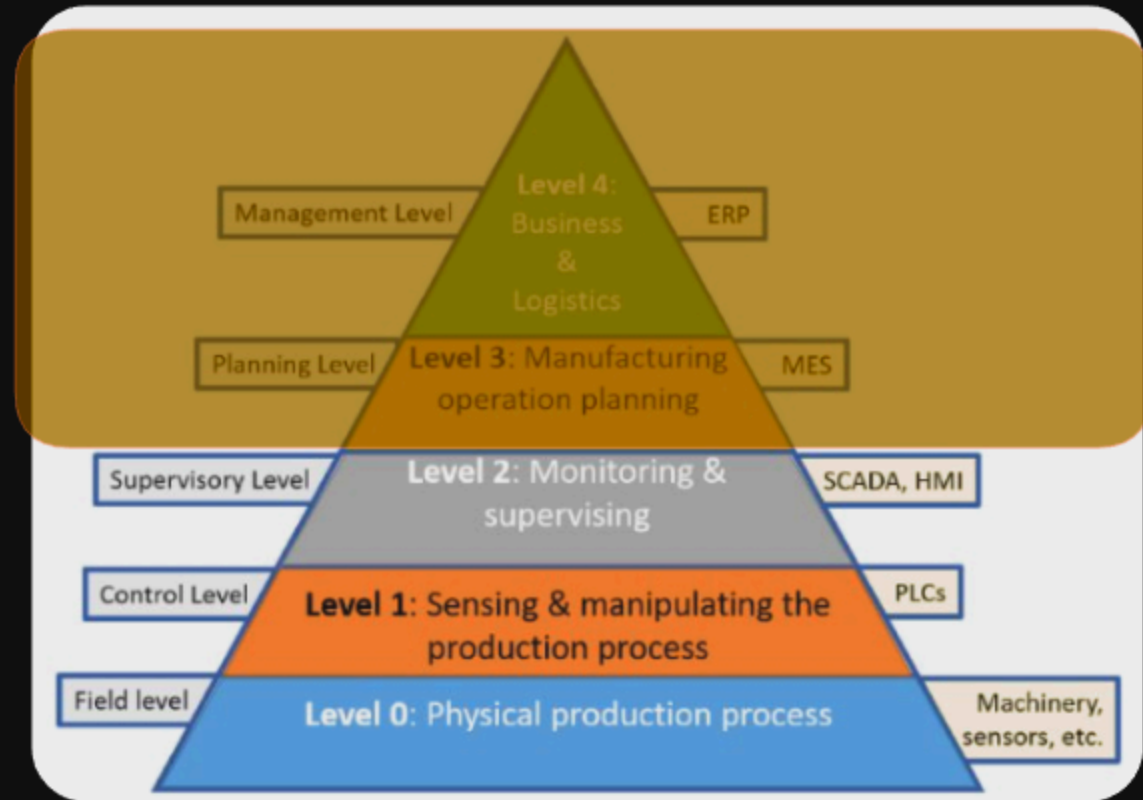
What are we doing in this course?

- We will use sensors and software that are related to these levels.
- I only pick the options that are accessible and affordable.
 - Open-source softwares
 - Low-cost sensors and devices
- But they are still relevant to real-world applications.





ERPNext



Class	Topic
1	ERPNext (Intro)
2	ERPNext (Sale, Purchase, Inventory)
3	ERPNext (Manufacturing)
4	Node-Red (Basic)
5	MQTT and Smart Sensors
6	Integration
7	Integration
8	Project

Let's go...